Journal of Pediatric Hematology/
Oncology Nursing
2024, Vol. 41(5) 309–314
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DOI: 10.1177/27527530231221145
journals.sagepub.com/home/jpo

Seroprevalence of Measles (Rubeola) Antibodies in Childhood Cancer Survivors

Deborah B. Crom, PhD, RN, CFNP[1], Lisa A. Walters, DNP, RN, CFNP[1], Yimei Li, PhD[2], Jai Liang, PhD[2], Diego R. Hijano, MD, MSc[3], Daniel A. Mulrooney, MD[4], Lee Ann Carmichael, DNP, RN, CFNP[1], Sarah L. Ford, MS, PA-C[1], Shekinah J. Andrews, MSN, RN, CFNP[1], Daniel Smith, DNP, RN, CFNP[1], Melissa M. Hudson, MD[4,5], and Belinda N. Mandrell, PhD, RN, PNP-AC, FAAN[6]

Abstract

Background: Measles is reemerging as a public health threat, raising important questions about disease vulnerability among childhood cancer survivors. This secondary analysis assessed the seroprevalence of anti-measles immunoglobulin G (IgG) antibodies as a marker of immune status in survivors of childhood cancer and associated demographic/treatment variables. **Method:** Participants were childhood cancer survivors who were free of active disease, having routine blood studies drawn, and could provide documentation of having received two doses of measles, mumps, and rubella vaccine before their cancer diagnosis. Patient record review documented demographic and treatment variables. Antimeasles (rubeola) IgG antibody seroprevalence was assessed by enzyme immunoassay for vaccine-specific antibodies. **Results:** Of 270 survivors evaluated, 110 (42%) were female, 196 (75%) were White, and 159 (61%) were leukemia/lymphoma survivors. Of these 262, 110 (42%) had negative measles seroprevalence, suggesting loss of immunity. **Conclusion:** Measles antibody surveillance and the need for reimmunization for survivors of childhood cancer survivors outside the transplant setting remains controversial. Our analysis indicates that a substantial proportion of survivors lose vaccine-related immunity to measles. Pediatric oncology nurses play important roles in educating cancer survivors regarding their risk of measles infection, evaluating the need for reimmunization, correcting misinformation about vaccine safety and effectiveness, and working to optimize community herd-based immunity.

Keywords

rubeola, seroprevalence, vaccine, childhood cancer survivors, cancer institute (NCI) and the American Lebanese Syrian associated charities

Introduction

Measles is a highly contagious infection spread by respiratory droplets. Approximately 90% of unvaccinated individuals exposed to the virus develop the disease. Serious complications associated with measles include pneumonia and encephalitis (Strebel & Orenstein, 2019). The introduction of vaccination against measles in 1963 dramatically reduced the incidence of the disease and the associated morbidity and mortality, and in 2000, the U.S. Centers for Disease Control and Prevention (U.S. CDC) declared that measles had been eliminated from the United States (Katz & Hinman, 2004). However, a November 2022 U.S. CDC Newsroom Release cautioned that measles poses a growing threat to nearly 40 million children as a consequence of routine immunization

[1]Center for Advanced Practice, St. Jude Children's Research Hospital, Memphis, TN, USA
[2]Department of Biostatistics, St. Jude Children's Research Hospital, Memphis, TN, USA
[3]Department of Infectious Diseases, St. Jude Children's Research Hospital, Memphis, TN, USA
[4]Department of Oncology, St. Jude Children's Research Hospital, Memphis, TN, USA
[5]Department of Epidemiology and Cancer Control, St. Jude Children's Research Hospital, Memphis, TN, USA
[6]Department of Pediatric Medicine, St. Jude Children's Research Hospital, Memphis, TN, USA

Corresponding Author:
Belinda Mandrell, PhD, RN, FAAN, St. Jude Children's Research Hospital, 262 Danny Thomas Place, MS 735, Memphis, TN 38105.
Email: Belinda.mandrell@stjude.org

programs having been disrupted during the COVID-19 pandemic, resulting in a steady decline in measles vaccination coverage (U.S. Centers for Disease Control and Prevention, 2022). The dramatic decrease in coverage is attributable not only to pandemic-related disruption but also to the rising culture of vaccine hesitancy, which has resulted in a loss of herd immunity (National Center for Immunization and Respiratory Diseases Do VD, 2019).

Childhood cancer survivors may be at a heightened risk for measles infection because of losing vaccine protection secondary to cancer therapy. The durability of measles immunity in childhood cancer survivors who receive cancer therapy after immunization is unknown. Several cohort studies have found a significant loss of measles vaccine–derived seroprotection, specifically in survivors of acute lymphoblastic leukemia (ALL; Bochennek et al., 2014). Other investigators exploring the serological status of survivors of childhood sarcoma found that 14% had negative measles seroprevalence (Yu et al., 2007). While another study reported over 75% of childhood cancer survivors who received two lifetime doses of the measles, mumps, and rubella (MMR) vaccine had a negative or equivocal titer to measles or mumps (Speckhart, 2023). A recent Children's Oncology Group review emphasized the importance of measles seroprevalence surveillance in bone marrow transplant recipients; however, the risk of infection in childhood cancer survivors treated with standard-risk therapy has not been fully addressed (Guilcher et al., 2021). Published studies of the serological status of childhood cancer survivors have been limited by the small sample sizes characteristic of single-institution studies and have yielded conflicting results. Another major limitation of these studies has been the lack of verified documentation that the participants had received two doses of MMR vaccine before their cancer was diagnosed.

The U.S. CDC recommends that children receive two doses of antimeasles vaccine as part of the MMR series. The first dose is given between 12 and 15 months of age and the second dose between 4 and 6 years of age. Two doses of vaccine confer measurable immunity in 94.1% of the immunized population (Haralambieva et al., 2019), and robust durability of measles immunity has been reported in healthy individuals up to 17 years after immunization (Kennedy et al., 2019). To eradicate measles, however, a herd immunity of at least 90 – 95% must be sustained (Patel et al., 2019).

The current culture of vaccine hesitancy places cancer survivors at increased risk of infection. Many vaccine-preventable diseases, such as measles, are now emerging because of the lack of community vaccine compliance. The highly contagious nature of measles, combined with the vulnerability of childhood cancer survivors in the context of vaccine undercoverage, highlights the need to increase community risk awareness. This secondary data analysis aimed to determine the measles antibody seroprevalence in childhood cancer survivors and serological factors. A better understanding of the measles seroprevalence will assist pediatric oncology nurses in counseling survivors regarding the need for reimmunization.

Method

Study Design

This secondary analysis was approved by the St. Jude Children's Research Hospital (St. Jude) Institutional Review Board. Because the study met the criteria for minimal risk, a waiver of consent was obtained.

Sample and Data Collection

A convenience sample of survivors attending the St. Jude After Completion of Therapy Clinic from January 2020 to February 2022 was identified. Survivors were eligible for inclusion if they were long-term survivors of leukemia, solid tumors, or brain tumors who were free of active disease, were having routine blood studies drawn, and could document receipt of two MMR vaccinations before commencing their cancer therapy. Long-term survivors treated with observation, resection alone, radiation therapy alone, or bone marrow transplantation were excluded. Clinical and demographic data were abstracted from the electronic medical records and included age, race, sex, primary cancer diagnosis, time from treatment, and time from the last MMR vaccination.

Two hundred and sixty-two survivors were included in the study. The median age of participants was 19 years, 58% were male, 75% were White, and 61% were survivors of leukemia or lymphoma (Table 1). Participants were a median of 8 years from their cancer diagnosis and a median of 14 years from their last MMR vaccination.

Laboratory Testing

Enzyme immunoassay (EIA) kits for detecting antimeasles immunoglobulin G (IgG) antibodies (Bio-Rad Laboratories) were used to detect and quantify antimeasles IgG antibodies in plasma samples. EIA microplates were processed using an EVOLIS System (Bio-Rad Laboratories). Tested samples were categorized as negative ($\leq$0.9 U/mL), equivocal (>0.9 to <1.1 U/mL), or positive ($\geq$1.1 U/mL) for the presence of antimeasles IgG antibodies reflecting immunity.

Statistical Analysis

The prevalence rate for this childhood cancer survivor cohort was defined as the proportion of patients with positive serology tests based on antibody detection. Patients with equivocal test results were excluded from the

Table 1 Sample Demographics and Clinical Characteristics

Total patients ($N = 262$)	
Age	
M (*SD*)	19.60 (3.91)
Mdn (min, max)	19.20 (11.6, 30.30)
Positive serology	
No	110 (42%)
Yes	152 (58%)
Sex	
Female	110 (42%)
Male	152 (58%)
Race	
White	196 (75%)
Black	50 (19%)
Other	16 (6%)
Diagnosis	
Leukemia/lymphoma	159 (61%)
Solid tumor	55 (21%)
Brain tumor	48 (18%)
Treatment	
Chemotherapy	169 (65%)
Chemotherapy + radiation	93 (35%)
Years from diagnosis	
M (*SD*)	7.92 (2.45)
Mdn (min, max)	7.50 (3, 18)
Years from vaccination	
M (*SD*)	14.40 (3.83)
Mdn (min, max)	14 (5, 25)

Table 2 Demographics According to Seroprevalence Rate

Seroprevalence rate (95% CI)	
Age in years	
< 20	0.53 (0.45–0.61)
≥ 20	0.64 (0.55–0.73)
Sex	
Female	0.64 (0.54–0.72)
Male	0.54 (0.46–0.62)
Race	
Black	0.74 (0.60–0.84)
Other	0.44 (0.23–0.67)
White	0.55 (0.48–0.62)
Diagnosis	
Brain tumor	0.54 (0.40–0.68)
Leukemia/lymphoma	0.55 (0.47–0.62)
Solid tumor	0.70 (0.58–0.81)
Treatment	
Chemotherapy	0.56 (0.49–0.63)
Chemotherapy + radiation	0.61 (0.51–0.70)

Note. CI = confidence interval.

Table 3 Prevalence Ratio Estimated by Univariate Poisson Regression

Univariate variable (excluding equivocal pts)	PR estimate (95% CI)	*p*
Age in years (≥20/<20)	1.18 (0.94–1.42)	.112
Sex		
Female/male	0.84 (0.67–1.01)	.100
Race		
Black/other	1.70 (0.72–2.69)	.071
	1.25 (0.54–1.98)	.429
Diagnosis		
Leukemia/brain tumor	0.99 (.70–1.28)	.957
Solid tumor/brain tumor	1.29 (0.89–1.68)	.108
Treatment		
Chemotherapy/chemotherapy + radiation	0.90 (0.71–1.09)	.321

Note. PR = prevalence ratio; CI = confidence interval.

analysis. Overall and subgroup prevalence rates were estimated with Wilson's binomial test under 95% confidence probability. Poisson multivariable regression with robust standard errors was used to estimate the prevalence ratio for each subgroup. Models were adjusted for age group (≥20 years vs < 20 years), race, sex, and diagnosis. SAS, Version 9.4 (SAS Institute) was used for analyses.

Results

Among the 262 study participants, 110 (42%) had negative seroprevalence for antimeasles IgG antibodies and 152 (58%) had positive seroprevalence for these antibodies. Seroprevalence estimates did not differ by diagnosis, race, age, sex, time since cancer diagnosis, or time since immunization (Table 2). Univariate modeling failed to identify statistically significant predictors of seroprevalence (Table 3).

Discussion

Only 58% of the childhood cancer survivors included in our study had positive seroprevalence for antimeasles IgG antibodies. This contrasts with the 92% seroprevalence measured in the general population in the United States, the 89% seroprevalence in Germany, and the 95% seroprevalence in Mexico (Diaz-Ortega et al., 2020; Friedrich et al., 2021; Lebo et al., 2015; McQuillan et al., 2007).

Although not statistically significant, the incidence of antimeasles antibody seroprevalence was higher in Black survivors than in other participants. This is consistent with reports that Black ancestry is associated with higher humoral and cellular lasting immune responses to measles vaccination (Voigt et al., 2016). In fact, in a survey of measles seropositivity rates in the United States from 1999 to 2004, the non-Hispanic Black population had the highest seroprevalence rate (McQuillan et al., 2007).

In previous studies sex and time since vaccination have been associated with measles serological status. However, we observed no significant difference between seropositive and seronegative study participants with respect to sex. Population studies have demonstrated that males and females differ in their humoral and cell-mediated responses to measles vaccines, with IgG antibody titers being significantly higher in females. Regression models indicate that age at the time of immunization and female sex are the two most significant predictors of antibody persistence after MMR vaccination (Mossong et al., 2000), but there is little consensus regarding the influence of sex on measles seroprevalence in childhood cancer survivors. Aytac et al. (2010) reported a significant sex-associated difference, with more male survivors than female survivors remaining seropositive for measles; however, other investigators reported no significant sex-based differences in seropositivity among survivors (Fayea et al., 2017; Fouda et al., 2018; Toret et al., 2021).

Likewise, time since vaccination did not influence seroprevalence in our cohort. Protection after MMR vaccination progressively declines over time, with vaccine-derived antibodies being less durable than those derived from infection. An evaluation of serum IgG titers in the general population at 7- and 17-years postvaccination documented a decrease of nearly 60% in protective titers (Kennedy et al., 2019).

We found no statistically significant difference in serological status based on cancer diagnosis, even though immune dysfunction has been widely documented in survivors of leukemia as having a lower seroprevalence for measles when compared with survivors of solid malignant neoplasms (Keskin Yildirim & Buyukavci, 2018). Other reports have confirmed that, compared to survivors of all other malignancies reviewed, survivors of ALL disproportionately lose protective antibody levels against measles (Bochennek et al., 2014; Fouda et al., 2018; Yu et al., 2007).

It is not yet well characterized how rapidly or at what point childhood cancer survivors lose seroprotection (Perkins et al., 2017). Protective measles antibodies were lacking in 42% of childhood cancer survivors in our cohort at a median of 8 years posttreatment. These findings highlight the risk posed by measles to childhood cancer survivors and the possible need for revaccination. However, the exact risk of measles in childhood cancer survivors is unknown because of herd immunity and the rarity of childhood cancer. Loss of herd immunity associated with increasing vaccine hesitancy in the general population raises concern for potentially serious clinical implications for childhood cancer survivors.

A recent review by the Children's Oncology Group endorses immune function testing after standard-dose chemotherapy only if there is a clinical suspicion of immune dysfunction. However, noting the lack of universal standards, the group defers the assessment of antibody titers in survivors to local providers. Decisions regarding re-immunization are then shared between survivors and providers on an individual basis (Guilcher et al., 2021; Robin et al., 2015).

In contrast, universal revaccination after pediatric cancer treatment has been advocated in the literature (Fouda et al., 2018). Current guidelines from the 2017 European Conference on Infectious Disease and Leukemia recommend administering one dose of MMR booster to anyone who received two doses before a diagnosis of cancer (irrespective of the antibody titers) as a therapeutically sound and cost-effective strategy to provide protection against measles. Serological follow-up of titers is noted to be expensive and not universally available. The guidelines conclude that revaccination represents an instrumental initiative to reduce morbidity and mortality in an era in which population vaccine coverage is low and declining (Mikulska et al., 2019).

The disparity between these recommendations is complicated by an increasing culture of vaccine hesitancy that heightens the threat of communicable diseases in childhood cancer survivors. Vaccine hesitancy is a multidimensional rubric of misinformation, convenience, and complacency (MacDonald & Hesitancy, 2015). Vaccine hesitancy and weak regulations requiring the vaccination of school-aged children place many communities below the protective thresholds of herd immunity, thereby contributing to recent outbreaks (Paules et al., 2019; Sanyaolu et al., 2019). The COVID-19 pandemic lockdown significantly decreased well-child vaccinations, further reducing critical herd immunity (Onimoe et al., 2022), and unvaccinated or undervaccinated individuals constituted a substantial portion of patients in recently reported measles outbreaks. The Internet has displaced medical clinicians as a trusted source of vaccine information, with social media platforms having been identified as the single most aggravating factor contributing to vaccine hesitancy. Their influential mode of spreading antivaccination information includes questioning scientific integrity, emotional appeals for civil liberties, and concerns regarding safety (Chung-Delgado et al., 2021).

Undergoing treatment for childhood cancer is a well-established risk factor for decreased immunity to measles, and this finding was replicated in our initiative. Evaluating pediatric cancer survivors for vaccine-preventable infectious diseases such as measles is an important part of supportive therapy. Although the guidelines for reimmunization after standard-dose chemotherapy remain unclear, focused consideration must be given to risk assessment/revaccination in vulnerable survivors.

Limitations

This study has several limitations. The relatively small sample size limited our analysis of potential associations between seroprevalence and clinical variables. No cohort of healthy controls was available for comparison. Survivors of leukemia accounted for 61% of the study participants, thereby limiting the generalizability of our results to other childhood cancer diagnoses. In addition, the study population may not be entirely representative of childhood cancer survivors, as some patients were hesitant to travel to our Survivorship Clinic during the COVID-19 pandemic.

Clinical Implications

Our results are a call to action for pediatric oncology nurses to assess survivors' risk of measles and to encourage repeat vaccination when appropriate. Not acting is no longer an option. Special attention must be given to young adult survivors living in close communities, such as colleges, trade schools, or the military, or working in healthcare settings where infectious diseases often emerge. Clinicians are in a pivotal position to address vaccine hesitancy both in their patients and in their communities. Interventions designed to address vaccine hesitancy stress the need to affirm patients' personal values while supplying information regarding the benefits of vaccinations. In addition, nurses can help curb the spread of misinformation by engaging in public forums and social media and encouraging focused surveillance and legislation to counteract erroneous information propagated via the Internet.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: Research Grant and Cancer Center Support Grant (CA21765) from the National Cancer Institute and ALSAC.

ORCID iD

Belinda N. Mandrell https://orcid.org/0000-0002-6041-7559

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